

D779

Software Product Design and Requirement Engineering

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A food-delivery app operating in 12 cities lists these requirements. Which one is non-functional?

- A. Order-status updates must reach the customer's phone within 3 seconds.
- B. The app launches in 12 cities.
- C. Customers can tip the driver in the app.
- D. Drivers can mark an order as delivered.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

Which quality attribute does this requirement express: "The system must be up 99.95% of each month"?

- A. Availability.
- B. It is a functional requirement.
- C. Usability.
- D. Performance.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

Warehouse pickers use many unwritten shortcuts, but in interviews they describe the official procedure instead of what they actually do. The warehouse has 180 pickers across three shifts. Which elicitation technique fits best?

- A. Reading the official procedure manual.
- B. Observation (job shadowing) on the floor.
- C. Longer interviews with the same pickers.
- D. A survey of all 180 pickers.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A clinic patient portal has a fixed launch date. Legal says a HIPAA-compliant login is required before launch; dark mode is popular but can wait; the sponsor loves an animated welcome screen. The team has six developers. How should the compliant login be classified?

- A. Could have.
- B. Should have.
- C. Won't have, because six developers are too few.
- D. Must have.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

A requirement is labeled “Won’t have (this time).” What does that mean?

- A. It must be built first.
- B. It is permanently rejected.
- C. It is agreed out of scope for this release but may return later.
- D. It will be built if time allows.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A document-management product serves 40 law firms. Which requirement is written best?

- A. The system shall be designed for 40 law firms.
- B. The system shall never fail.
- C. The system shall be fast and user-friendly.
- D. Search results for a client name shall display within 2 seconds for 95% of queries.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

Which rewrite makes “The report shall load quickly” testable?

- A. Users shall be satisfied with load speed.
- B. The monthly report shall load within 5 seconds for datasets up to 1 million rows.
- C. The report shall load very quickly.
- D. The report shall always load instantly.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A gym chain with 30 locations is writing its backlog. Which is a well-formed user story?

- A. As a user, I want the app to be good.
- B. As a member, I want to book a class from my phone so I can secure a spot. Done when: booking is confirmed within 10 seconds, and full classes offer a waitlist.
- C. As the company, I want all 30 locations listed.
- D. Set up the front-end framework and database schema.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A hospital is buying a nurse-scheduling system. The CFO signs the contract, nurses will use it every shift, IT will maintain it, and the union contract sets shift rules. The vendor is based in Ohio. Whose daily tasks should drive the functional requirements?

- A. The vendor in Ohio.
- B. The union.
- C. The CFO.
- D. The nurses.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A product team rates four features. A: value 40, cost 10. B: value 60, cost 20. C: value 30, cost 5. D: value 80, cost 40. Feature A was requested first by the sales VP. The team plans on a digital whiteboard. Ranking by value-to-cost ratio, which feature goes first?

- A. Feature D, with the highest value.
- B. Feature B or D, which tie for the highest value minus cost.
- C. Feature C.
- D. Feature A, because it was requested first.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

Feature E has a value of 72 and needs a value-to-cost ratio of at least 4 to make the release. What is the most it can cost?

- A.** 288
- B.** 68
- C.** 18
- D.** 4

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

Stakeholders signed off on a 90-page requirements document written by two senior analysts. At the first demo, they say the dashboard is “not what we meant.” What should have been done earlier?

- A.** Review clickable prototypes or mockups of the dashboard with stakeholders before building.
- B.** Write a longer, more detailed document.
- C.** Assign more senior analysts.
- D.** Freeze requirements and allow no changes.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

A food-delivery app ~~operating in 12 cities~~ lists these requirements. Which one is non-functional?

Target word: identify non-functional **Decides:** A measurable quality, not a behavior.

A	Correct	A performance requirement.
B	Camouflage	A business rollout plan, not a system quality.
C	Inverted operation	A behavior: functional.
D	Confusion set	Also a behavior.

2. Answer A Concept 1 • transfer

Which quality attribute does this requirement express: “The system must be up 99.95% of each month”?

Target word: name the attribute — reversed **Decides:** Uptime percentage.

A	Correct	Measures the share of time the system is usable.
B	Inverted operation	It states a quality, not a behavior.
C	Confusion set	Ease of use, not uptime.
D	Adjacent category	Speed and throughput, not uptime.

3. Answer B Concept 2

Warehouse pickers use many unwritten shortcuts, but in interviews they describe the official procedure instead of what they actually do. ~~The warehouse has 180 pickers across three shifts.~~ Which elicitation technique fits best?

Target word: technique choice **Decides:** Real practice is tacit and not reported in interviews.

A	Adjacent category	Documents the stated process, not the real one.
B	Correct	Shows what people actually do.
C	Dropped element	More of the method that is already failing.
D	Camouflage	Reach does not fix self-reporting of the official procedure.

4. Answer D Concept 3

A clinic patient portal has a fixed launch date. Legal says a HIPAA-compliant login is required before launch; dark mode is popular but can wait; the sponsor loves an animated welcome screen. ~~The team has six developers.~~ How should the compliant login be classified?

Target word: MoSCoW category **Decides:** Launch cannot legally happen without it.

A	Confusion set	Could fits nice-to-haves like the animation.
B	Adjacent category	Should items are important but not launch-blocking.
C	Camouflage	Team size does not change a legal requirement.
D	Correct	The release fails without it.

5. Answer C Concept 3 · transfer

A requirement is labeled “Won’t have (this time).” What does that mean?

Target word: meaning — reversed from label **Decides:** Out of scope for this release only.

A	Inverted operation	That is Must have.
B	Absolute language	“This time” limits it to the current release.
C	Correct	Won’t sets expectations without rejecting forever.
D	Adjacent category	That is Could have.

6. Answer D Concept 4

A document-management product ~~serves 40 law firms~~. Which requirement is written best?

Target word: best-written requirement **Decides:** Measurable and testable.

A	Camouflage	Echoes a stem fact but states no testable behavior.
B	Absolute language	Impossible to meet or test.
C	Dropped element	No measure.
D	Correct	A tester can pass or fail it.

7. Answer B Concept 4 · transfer

Which rewrite makes “The report shall load quickly” testable?

Target word: fix an untestable requirement **Decides:** Needs a number, a condition, and a scope.

A	Adjacent category	Subjective, not a measurable speed.
B	Correct	Measurable threshold and conditions.
C	Dropped element	Still no measure.
D	Absolute language	Not achievable or testable.

8. Answer B Concept 5

A gym chain ~~with 30 locations~~ is writing its backlog. Which is a well-formed user story?

Target word: well-formed story **Decides:** Role, goal, benefit, and acceptance criteria.

A	Dropped element	No specific goal, benefit, or criteria.
B	Correct	Has role, goal, benefit, and testable criteria.
C	Camouflage	Echoes a stem fact; no user benefit or criteria.
D	Adjacent category	A technical task, not user value.

9. Answer D Concept 6

A hospital is buying a nurse-scheduling system. The CFO signs the contract, nurses will use it every shift, IT will maintain it, and the union contract sets shift rules. ~~The vendor is based in Ohio.~~ Whose daily tasks should drive the functional requirements?

Target word: primary end users **Decides:** Nurses use it every shift.

A	Camouflage	The supplier’s location does not define needs.
B	Adjacent category	Its contract adds constraints, not daily tasks.
C	Confusion set	The sponsor or customer funds it but does not use it daily.
D	Correct	End users’ tasks define what the system must do.

10. Answer C Concept 7

A product team rates four features. A: value 40, cost 10. B: value 60, cost 20. C: value 30, cost 5. D: value 80, cost 40. ~~Feature A was requested first by the sales VP. The team plans on a digital whiteboard.~~ Ranking by value-to-cost ratio, which feature goes first?

Target word: top feature by ratio **Decides:** Ratios: A 4, B 3, C 6, D 2.

A	Dropped element	Drops cost; D has the lowest ratio (2).
B	Adjacent category	Net value (40 each) is a different measure than the ratio.
C	Correct	$30 / 5 = 6$, the highest ratio.
D	Camouflage	Request order is not in the ratio.

11. Answer C Concept 7 · transfer

Feature E has a value of 72 and needs a value-to-cost ratio of at least 4 to make the release. What is the most it can cost?

Target word: solve for cost **Decides:** Cost = value ÷ ratio.

A	Inverted operation	Multiplies instead of dividing.
B	Adjacent category	Subtracts instead of dividing.
C	Correct	$72 / 4 = 18$.
D	Wrong stem number	The required ratio.

12. Answer A Concept 8

Stakeholders signed off on a 90-page requirements document ~~written by two senior analysts.~~ At the first demo, they say the dashboard is “not what we meant.” What should have been done earlier?

Target word: validation practice **Decides:** Misunderstanding surfaced only at demo.

A	Correct	Prototypes reveal mismatches early and cheaply.
B	Dropped element	More text still skips a hands-on check.
C	Camouflage	Seniority does not decide validation method.
D	Absolute language	Locks in the misunderstanding.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	6	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	9	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	4	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	4	
Camouflage	Answers a stem fact that was never the question.	8	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Functional vs. non-functional requirements	Functional requirements describe behaviors the system performs; non-functional requirements describe qualities such as performance, availability, security, and usability.	Classifying a feature as a quality, or a quality as a feature.	“Which is non-functional?” or a percentage uptime or response time.
2	Choosing an elicitation technique	Interviews capture stated views; observation captures tacit, actual practice; surveys reach many people with fixed questions; workshops build consensus.	Relying on interviews when people cannot describe what they really do.	Users describe the official procedure but work differently.
3	MoSCoW prioritization	Must = release fails without it; Should = important but not vital; Could = nice if time allows; Won’t = agreed out of scope for this release.	Treating Won’t as rejected forever, or legal needs as Should.	A fixed release date and a mix of mandatory and nice-to-have items.
4	Testable, unambiguous requirements	A good requirement states a measurable condition that a test can pass or fail.	Words like fast, easy, always, never without a measure.	“Which requirement is best written?”
5	User stories and acceptance criteria	“As a [role], I want [goal] so that [benefit],” plus acceptance criteria that define done.	Writing technical tasks or vague wishes as stories.	Several candidate stories to choose from.
6	Stakeholder roles	End users drive task-level functional needs; sponsors or customers fund and approve; regulators and contracts impose constraints.	Treating the buyer as the user.	A list of parties around a system purchase.
7	Value-to-cost prioritization	Rank features by value ÷ cost to get the most value per unit of effort.	Ranking by value alone or by value minus cost.	A table with value points and cost points.
8	Requirements validation with prototypes	Prototypes and walkthroughs expose misunderstandings early, before build.	Relying on long signed documents alone.	Stakeholders signed off, then said “that’s not what we meant.”

Course scope (public description): Software Product Design and Requirement Engineering provides an in-depth exploration into effectively integrating user needs and system requirements into software development. This course covers topics from types of requirements, requirement gathering techniques, prioritization, and documentation.